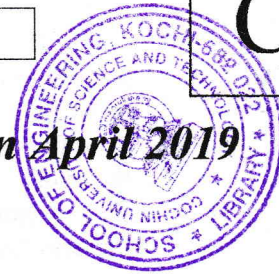


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# **B.Tech. Degree IV Semester Examination April 2019**

## **ME 1404 APPLIED THERMODYNAMICS (2012 Scheme)**



Time: 3 Hours

Maximum Marks: 100

### **PART A** (Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Explain Clausius Clapeyron equations.
- (b) Explain the terms: irreversibility, exergy and dead state.
- (c) Explain P-v, P-T and h-s diagram of a pure substance which expands on freezing.
- (d) Explain the working of binary vapor cycle with a neat sketch.
- (e) Derive an expression for the pressure ratio for maximum discharge through a convergent-divergent nozzle.
- (f) Why compounding is done in steam turbines? Differentiate between pressure compounding and velocity compounding of steam turbines with neat sketches.
- (g) Differentiate between gravimetric analysis and volumetric analysis.
- (h) Derive the equation  $P_i/P = n_i/n = V_i/V$  for gas mixtures.

### **PART B**

(4 × 15 = 60)

- II. (a) A mass of 6.98 kg of air is in a vessel at 200 kPa, 27°C. Heat is transferred to the air from a reservoir at 727°C until the temperature of air rises to 327°C. The environment is at 100 kPa, 17°C. Determine (i) the initial and final availability of air (ii) maximum useful work associated with the process. (9)
- (b) Establish the equivalence of Kelvin-Planck and Clausius statements. (6)

**OR**

- III. A heat pump working on the Carnot cycle takes in heat from a reservoir at 5°C and delivers heat to a reservoir at 60°C. The heat pump is driven by a reversible heat engine which takes heat from a reservoir at 840°C and rejects heat to a reservoir at 60°C. The reversible heat engine also drives a machine that absorbs 30 kW. If the heat pump extracts 17 kW from 5°C reservoir, determine (i) rate of heat supply from 840°C source (ii) rate of heat rejection to 60°C sink. (15)

- IV. In a Rankine cycle, the steam at inlet to the turbine is saturated at a pressure of 35 bar and the exhaust pressure is 0.2 bar. Assuming a flow rate of 9.5 kg/s determine: (15)

- (i) The pump work
- (ii) The turbine work
- (iii) The Rankine efficiency
- (iv) The condenser heat flow
- (v) The dryness at the end of expansion

**OR****(P.T.O.)**

- V. A turbine is supplied with steam at a pressure of 32 bar and a temperature of  $410^{\circ}\text{C}$ . The steam then expands isentropically to a pressure of 0.08 bar. Find the dryness fraction at the end of expansion and thermal efficiency of the cycle. If the steam is reheated at 5.5 bar to a temperature of  $395^{\circ}\text{C}$  and then expanded isentropically to a pressure of 0.08 bar, what will be the dryness fraction and thermal efficiency of the cycle? (15)

- VI. The velocity of steam entering the nozzle of an impulse turbine is 400 m/s. The blades operate close to the maximum blade efficiency. The nozzle angle is  $20^{\circ}$ . Considering equiangular blades and neglecting blade friction, calculate for a steam flow of 0.6 kg/s, the diagram power and diagram efficiency. (15)

OR

- VII. Steam at a pressure of 15 bar and dryness fraction of 0.97 is discharged through a convergent-divergent nozzle to a back pressure of 0.2 bar. The mass flow rate is 9 kg/kWhr. If the power developed is 20 kW, determine: (15)

- Throat pressure
- Number of nozzles required, if each nozzle has a throat of rectangular cross-section of  $4\text{ mm} \times 6\text{ mm}$ .
- If 15% of the overall isentropic enthalpy drop reheats the steam by friction in the divergent portion, find the cross-section of the exit rectangle.

- VIII. (a) A vessel of  $6\text{ m}^3$  capacity contains two gases A and B in proportion of 45% and 55% respectively at  $30^{\circ}\text{C}$ . If the value of  $R$  for the gases is  $0.288\text{ kJ/kg K}$  and  $0.295\text{ kJ/kg K}$  and if the total mass of the mixture is 2 kg, calculate: (7)

- The partial pressure
- The total pressure
- The mean value of  $R$  for the mixture

- (b) Air ( $\text{N}_2 = 77\%$ ,  $\text{O}_2 = 23\%$  by weight) at  $25^{\circ}\text{C}$  and 12 bar is contained in a vessel of capacity  $0.6\text{ m}^3$ . Some quantity of  $\text{CO}_2$  is forced into the vessel so that the temperature remains at  $25^{\circ}\text{C}$ ; but the pressure rises to 18 bar. Find the masses of  $\text{O}_2$ ,  $\text{N}_2$  and  $\text{CO}_2$  in the cylinder. (8)

OR

- IX. (a) Following is the gravimetric analysis of air (8)

| Constituent    | Percentage |
|----------------|------------|
| Oxygen         | 23.14      |
| Nitrogen       | 75.53      |
| Argon          | 1.28       |
| Carbon dioxide | 0.005      |

Calculate the analysis by volume and partial pressure of each constituent when the total pressure is 1 bar.

- (b) A vessel of  $3\text{ m}^3$  capacity contains a mixture of nitrogen and carbon monoxide, the analysis by volume showing quantities of each. The temperature is  $15^{\circ}\text{C}$  and total pressure is 3.5 bar. Determine the mass of each constituent. (7)

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